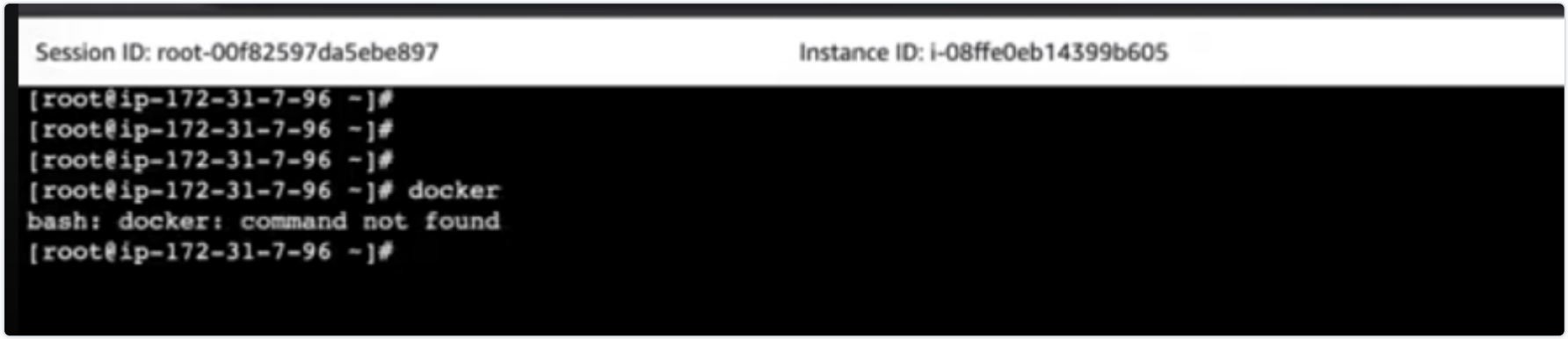
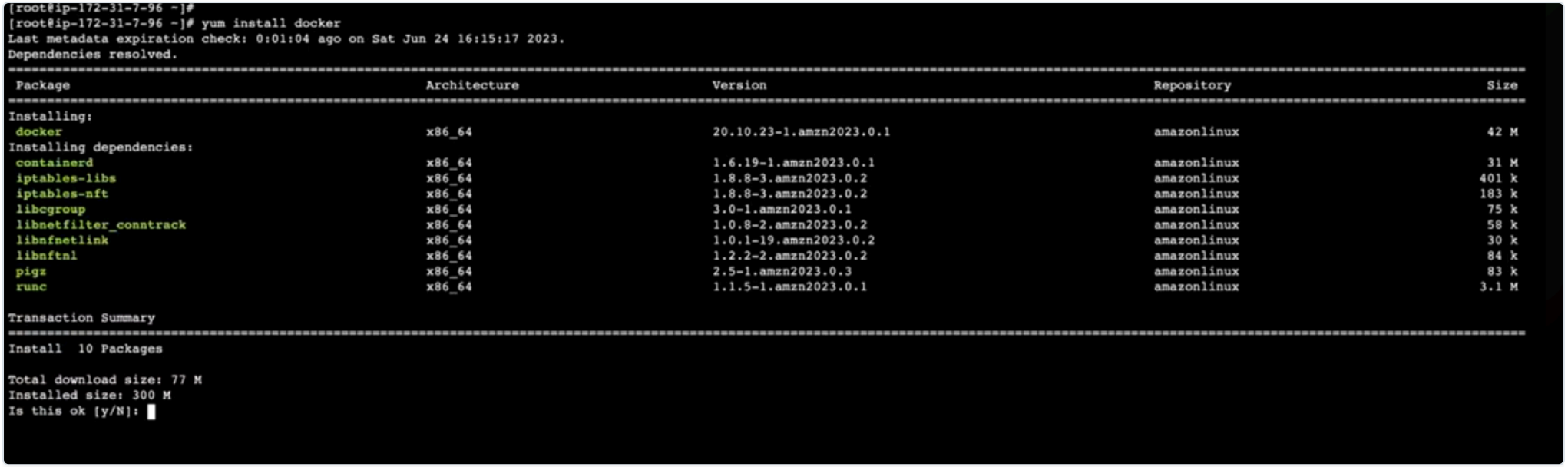
Screenshots — Docker Internals Hands-On (dockerd, containerd, shim, runc, /proc)

Kubernetes Lecture 12 · 32 shots, chronological · source stills from the session (2026-10-08 14:49 – 15:53)

01 / 32 Fresh EC2 (SSM session) — `docker: command not found` 14:49:08



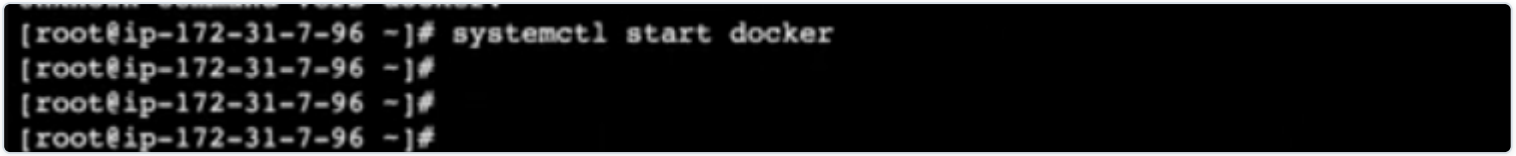

[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# yum install docker
Last metadata expiration check: 0:01:04 ago on Sat Jun 24 16:15:17 2023.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
docker x86_64 20.10.23-1.amzn2023.0.1 amazonlinux 42 M
Installing dependencies:
containerd x86_64 1.6.19-1.amzn2023.0.1 amazonlinux 31 M
iptables-libs x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 401 k
iptables-nft x86_64 1.8.8-3.amzn2023.0.2 amazonlinux 183 k
libcgroup x86_64 3.0-1.amzn2023.0.1 amazonlinux 75 k
libnetfilter_conntrack x86_64 1.0.8-2.amzn2023.0.2 amazonlinux 58 k
libnfnetlink x86_64 1.0.1-19.amzn2023.0.2 amazonlinux 30 k
libnftnl x86_64 1.2.2-2.amzn2023.0.2 amazonlinux 84 k
pigz x86_64 2.5-1.amzn2023.0.3 amazonlinux 83 k
runc x86_64 1.1.5-1.amzn2023.0.1 amazonlinux 3.1 M
Transaction Summary
Install 10 Packages
Total download size: 77 M
Installed size: 300 M
Is this ok [y/N]:

```
Installed:
  containerd-1.6.19-1.amzn2023.0.1.x86_64      docker-20.10.23-1.amzn2023.0.1.x86_64              iptables-libs-1.8.8-3.amzn2023.0.2.x86_64      iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
  libcgroup-3.0-1.amzn2023.0.1.x86_64          libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64     libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.3.x86_64               runc-1.1.5-1.amzn2023.0.1.x86_64
```

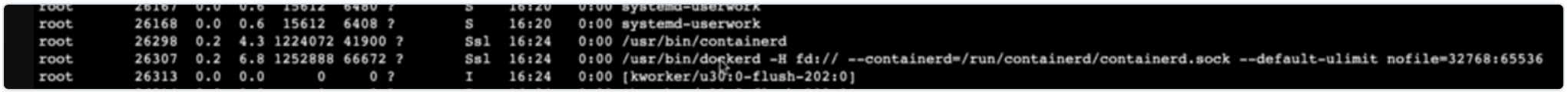
```
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# docker ps
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
[root@ip-172-31-7-96 ~]#
```

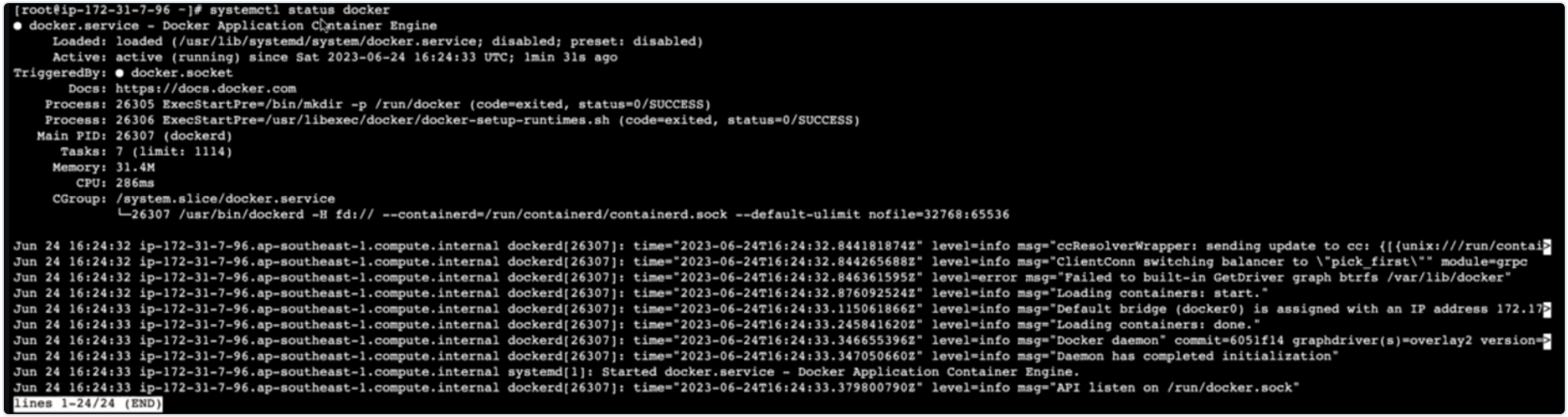
`ss -tnlp` — only sshd listening, no docker / containerd yet


```
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# ss -tnlp
State               Recv-Q              Send-Q                           Local Address:Port                           Peer Address:Port              Process
LISTEN              0                   128                                    0.0.0.0:22                                  0.0.0.0:*                  users:(("sshd",pid=2042,fd=5))
LISTEN              0                   128                                       [::]:22                                     [::]:*                  users:(("sshd",pid=2042,fd=7))
```

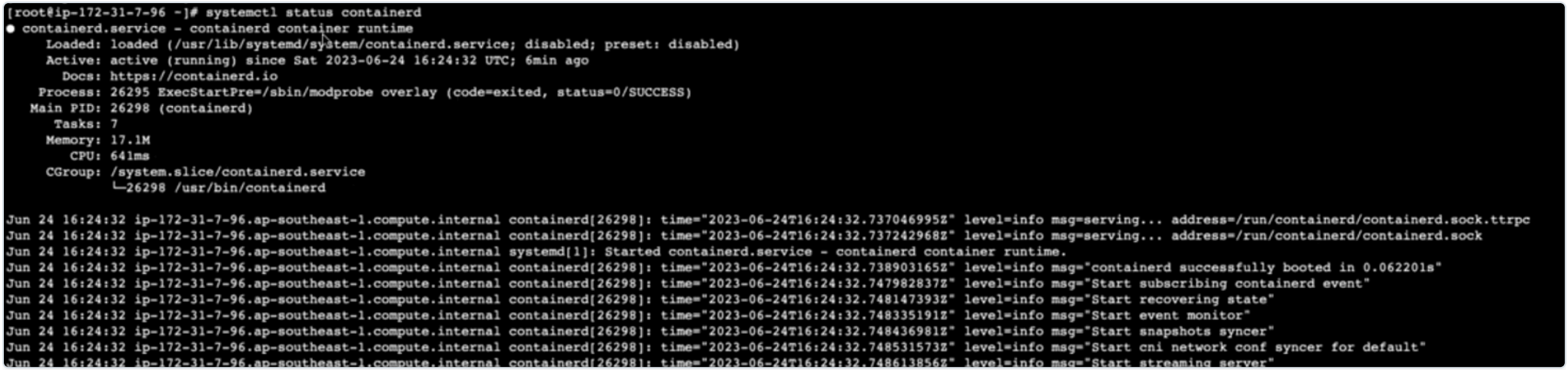

[root@ip-172-31-7-96 ~]# systemctl start docker
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]#


[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
[root@ip-172-31-7-96 ~]#


root 26168 0.0 0.6 15612 6408 ? S 16:20 0:00 systemd-userwork
root 26298 0.2 4.3 1224072 41900 ? Ssl 16:24 0:00 /usr/bin/containerd
root 26307 0.2 6.8 1252888 66672 ? Ssl 16:24 0:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536
root 26313 0.0 0.0 0 0 ? I 16:24 0:00 [kworker/u30:0-flush-202:0]


[root@ip-172-31-7-96 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
Active: active (running) since Sat 2023-06-24 16:24:33 UTC; 1min 31s ago
TriggeredBy: ● docker.socket
Docs: https://docs.docker.com
Process: 26305 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
Process: 26306 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
Main PID: 26307 (dockerd)
Tasks: 7 (limit: 1114)
Memory: 31.4M
CPU: 286ms
CGroup: /system.slice/docker.service
└─26307 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:32.844181874Z" level=info msg="ccResolverWrapper: sending update to cc: {[{unix:///run/contai
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:32.844265688Z" level=info msg="ClientConn switching balancer to \"pick_first\"" module=grpc
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:32.846361595Z" level=error msg="Failed to built-in GetDriver graph btrfs /var/lib/docker"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:32.876092524Z" level=info msg="Loading containers: start."
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:33.115061866Z" level=info msg="Default bridge (docker0) is assigned with an IP address 172.17
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:33.245841620Z" level=info msg="Loading containers: done."
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:33.346655396Z" level=info msg="Docker daemon" commit=6051f14 graphdriver(s)=overlay2 version=
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:33.347050660Z" level=info msg="Daemon has completed initialization"
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
Jun 24 16:24:33 ip-172-31-7-96.ap-southeast-1.compute.internal dockerd[26307]: time="2023-06-24T16:24:33.379800790Z" level=info msg="API listen on /run/docker.sock"
lines 1-24/24 (END)

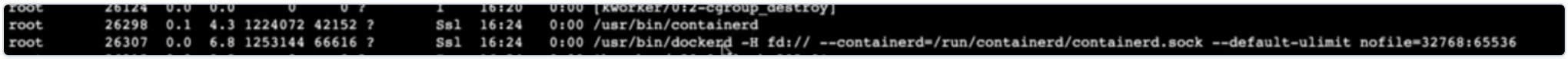
```
     CGroup: /system.slice/docker.service
             └─26307 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536
```



```
[root@ip-172-31-7-96 ~]# systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; disabled; preset: disabled)
     Active: active (running) since Sat 2023-06-24 16:24:32 UTC; 6min ago
       Docs: https://containerd.io
    Process: 26295 ExecStartPre=/sbin/modprobe overlay (code=exited, status=0/SUCCESS)
   Main PID: 26298 (containerd)
      Tasks: 7
     Memory: 17.1M
        CPU: 641ms
     CGroup: /system.slice/containerd.service
             └─26298 /usr/bin/containerd

Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.737046995Z" level=info msg=serving... address=/run/containerd/containerd.sock.ttrpc
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.737242968Z" level=info msg=serving... address=/run/containerd/containerd.sock
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal systemd[1]: Started containerd.service - containerd container runtime.
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.738903165Z" level=info msg="containerd successfully booted in 0.062201s"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.747982837Z" level=info msg="Start subscribing containerd event"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.748147393Z" level=info msg="Start recovering state"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.748335191Z" level=info msg="Start event monitor"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.748436981Z" level=info msg="Start snapshots syncer"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.748531573Z" level=info msg="Start cni network conf syncer for default"
Jun 24 16:24:32 ip-172-31-7-96.ap-southeast-1.compute.internal containerd[26298]: time="2023-06-24T16:24:32.748613856Z" level=info msg="Start streaming server"
```

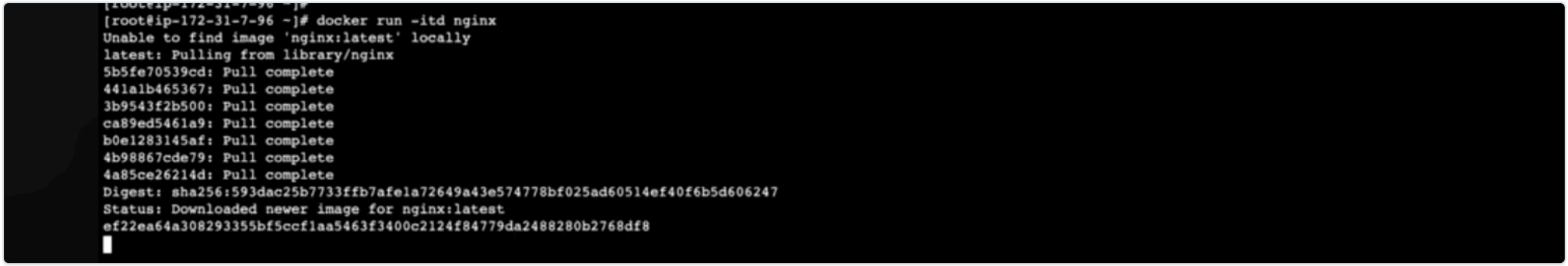
ps aux — containerd and dockerd both running



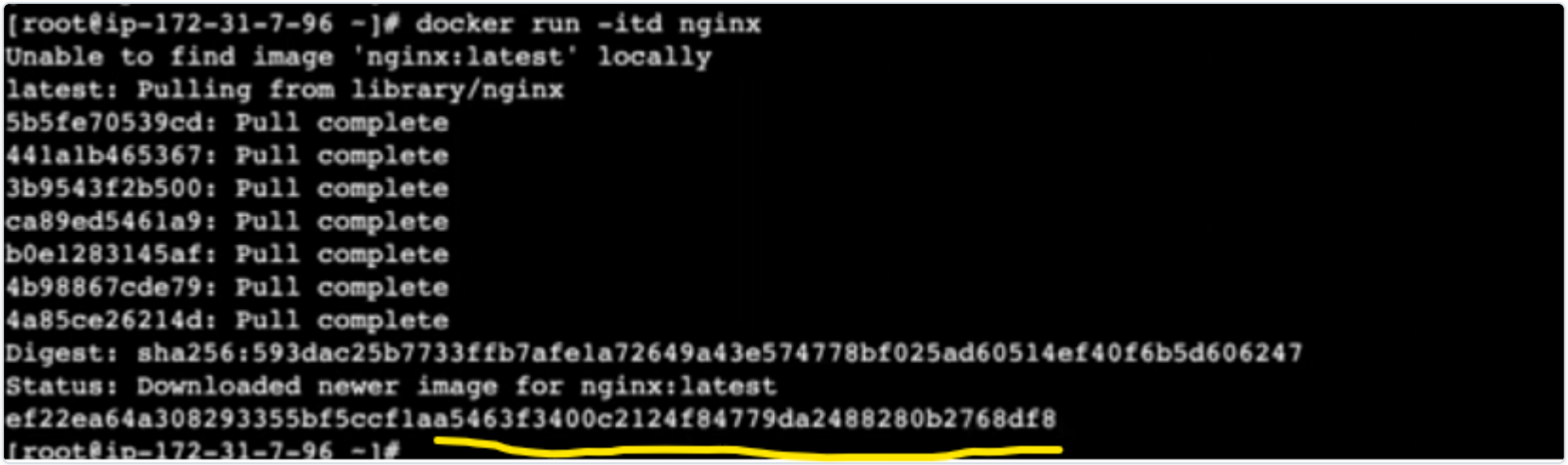
`ss -tnlp` — containerd listening on 127.0.0.1:41841 (CRI streaming server)



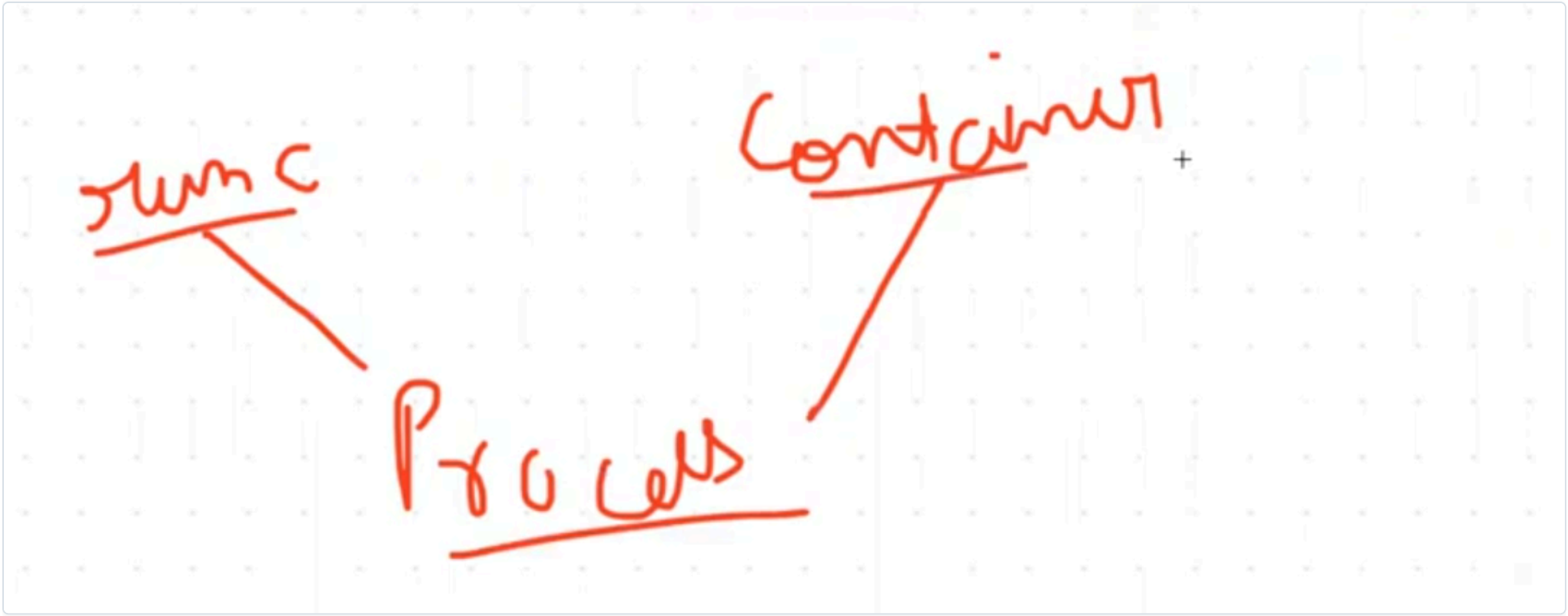

bash: docker: command not found
[root@ip-172-31-7-96 ~]# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
[root@ip-172-31-7-96 ~]#


[root@ip-172-31-7-96 ~]# docker run -itd nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
5b5fe70539cd: Pull complete
441a1b465367: Pull complete
3b9543f2b500: Pull complete
ca89ed5461a9: Pull complete
b0e1283145af: Pull complete
4b98867cde79: Pull complete
4a85ce26214d: Pull complete
Digest: sha256:593dac25b7733ffb7afe1a72649a43e574778bf025ad60514ef40f6b5d606247
Status: Downloaded newer image for nginx:latest
ef22ea64a308293355bf5ccf1aa5463f3400c2124f84779da2488280b2768df8

Same run — the unique container ID (SHA) underlined



Whiteboard — runc → PROCESS ← container



```
root       26625  0.0  0.0      0     0 ?        I    16:30   0:00 [kworker/u30:2-flush-202:0]
root       26626  0.0  0.0      0     0 ?        I    16:30   0:00 [kworker/u30:3]
root       27017  0.0  1.4 1356788 13824 ?       Sl   16:34   0:00 /usr/bin/containerd-shim-runc-v2 -namespace moby -id ef22ea64a308293355bf5ccf1aa5463f3400c2124f84779da2488280b2768df8 -address /r
```

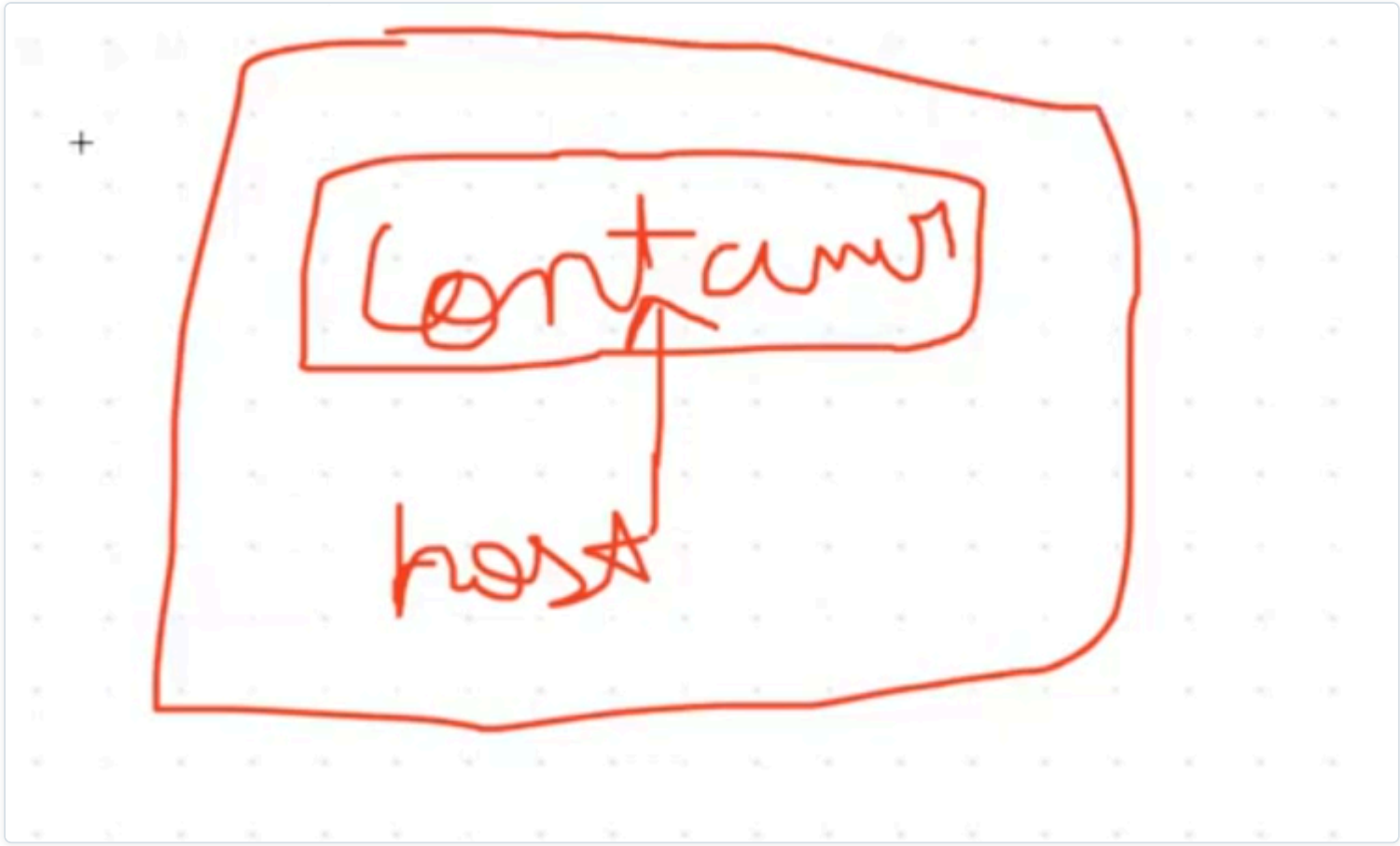
Shim command close-up — `-namespace moby`, `-id` = container ID


```
root       27017  0.0  1.4 1356788 13824 ?       Sl   16:34   0:00 /usr/bin/containerd-shim-runc-v2 -namespace moby -id ef22ea64a308293355bf5ccf1aa5463f3400c212
```

Whiteboard — full shim command incl. `-address /run/containerd/containerd.sock`



Whiteboard — the container takes its resources from the host

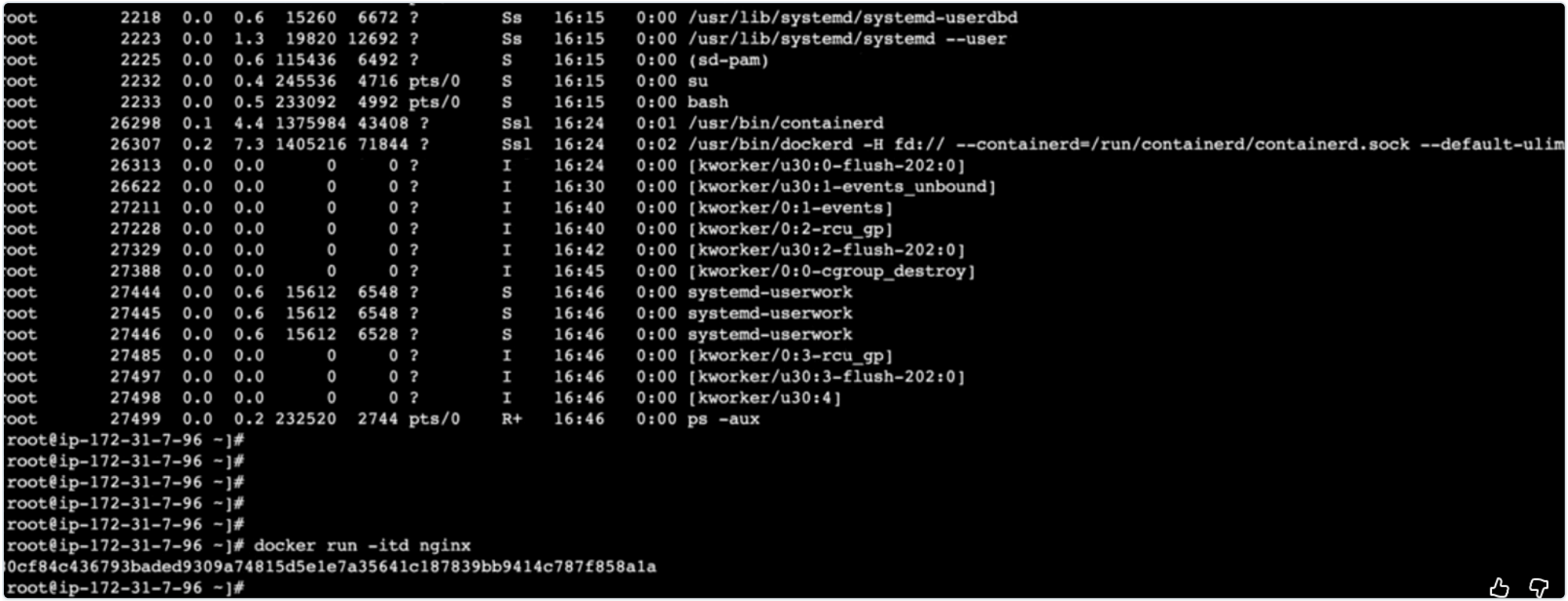


Shim process 27017 still running


```
root       27017  0.0  1.4 1356788 14024 ?       Sl   16:34   0:00 /usr/bin/containerd-shim-runc-v2 -namespace moby -id ef22ea64a308293355bf5ccf1aa5463f3400c212
```

```
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS     NAMES
ef22ea64a308   nginx     "/docker-entrypoint.…"   11 minutes ago   Up 11 minutes   80/tcp    adoring_ishizaka
```

```
[root@ip-172-31-7-96 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS     NAMES
ef22ea64a308   nginx     "/docker-entrypoint.…"   11 minutes ago   Up 11 minutes   80/tcp    adoring_ishizaka
[root@ip-172-31-7-96 ~]# docker kill ef22
ef22
```


oot 2218 0.0 0.6 15260 6672 ? Ss 16:15 0:00 /usr/lib/systemd/systemd-userdbd
oot 2223 0.0 1.3 19820 12692 ? Ss 16:15 0:00 /usr/lib/systemd/systemd --user
oot 2225 0.0 0.6 115436 6492 ? S 16:15 0:00 (sd-pam)
oot 2232 0.0 0.4 245536 4716 pts/0 S 16:15 0:00 su
oot 2233 0.0 0.5 233092 4992 pts/0 S 16:15 0:00 bash
oot 26298 0.1 4.4 1375984 43408 ? Ssl 16:24 0:01 /usr/bin/containerd
oot 26307 0.2 7.3 1405216 71844 ? Ssl 16:24 0:02 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulim
oot 26313 0.0 0.0 0 0 ? I 16:24 0:00 [kworker/u30:0-flush-202:0]
oot 26622 0.0 0.0 0 0 ? I 16:30 0:00 [kworker/u30:1-events_unbound]
oot 27211 0.0 0.0 0 0 ? I 16:40 0:00 [kworker/0:1-events]
oot 27228 0.0 0.0 0 0 ? I 16:40 0:00 [kworker/0:2-rcu_gp]
oot 27329 0.0 0.0 0 0 ? I 16:42 0:00 [kworker/u30:2-flush-202:0]
oot 27388 0.0 0.0 0 0 ? I 16:45 0:00 [kworker/0:0-cgroup_destroy]
oot 27444 0.0 0.6 15612 6548 ? S 16:46 0:00 systemd-userwork
oot 27445 0.0 0.6 15612 6548 ? S 16:46 0:00 systemd-userwork
oot 27446 0.0 0.6 15612 6528 ? S 16:46 0:00 systemd-userwork
oot 27485 0.0 0.0 0 0 ? I 16:46 0:00 [kworker/0:3-rcu_gp]
oot 27497 0.0 0.0 0 0 ? I 16:46 0:00 [kworker/u30:3-flush-202:0]
oot 27498 0.0 0.0 0 0 ? I 16:46 0:00 [kworker/u30:4]
oot 27499 0.0 0.2 232520 2744 pts/0 R+ 16:46 0:00 ps -aux
root@ip-172-31-7-96 ~]#
root@ip-172-31-7-96 ~]#
root@ip-172-31-7-96 ~]#
root@ip-172-31-7-96 ~]#
root@ip-172-31-7-96 ~]#
root@ip-172-31-7-96 ~]# docker run -itd nginx
0cf84c436793baded9309a74815d5e1e7a35641c187839bb9414c787f858a1a
root@ip-172-31-7-96 ~]#

New shim 27641 + nginx master 27664 (root) + worker 27714 (uid 101)


```
root       27485  0.0  0.0      0     0 ?        I    16:46   0:00 [kworker/0:3-xfs-inodegc/xvda1]
root       27497  0.0  0.0      0     0 ?        I    16:46   0:00 [kworker/u30:3-flush-202:0]
root       27498  0.0  0.0      0     0 ?        I    16:46   0:00 [kworker/u30:4-flush-202:0]
root       27641  0.2  1.2 1356468 12460 ?       Sl   16:47   0:00 /usr/bin/containerd-shim-runc-v2 -namespace moby -id 80cf84c436793baded9309a74815d5e1e7a35641
root       27664  0.7  0.7  11376  7360 pts/0    Ss+  16:47   0:00 nginx: master process nginx -g daemon off;
101        27714  0.0  0.2  11836  2764 pts/0    S+   16:47   0:00 nginx: worker process
```

```
[root@ip-172-31-7-96 ~]# cd
[root@ip-172-31-7-96 ~]# cd /
[root@ip-172-31-7-96 /]# cd proc
[root@ip-172-31-7-96 proc]#
[root@ip-172-31-7-96 proc]#
[root@ip-172-31-7-96 proc]#
[root@ip-172-31-7-96 proc]# ls
1     15    1713  1765  1837  2078  2233   27329  27714  35  79   975        devices        ioports      latency_stats  pagetypeinfo  sys            xen
10    157   1714  1767  1839  21    23     27388  27718  36  8    976        diskstats      irq          loadavg        partitions    sysrq-trigger  zoneinfo
1040  158   1718  1790  2     2193  24     27444  28     37  93   acpi       dma            kallsyms     locks          pressure      sysvipc
11    16    1719  18    20    22    26298  27445  29     38  94   buddyinfo  driver         kcore        mdstat         schedstat     thread-self
12    1666  1720  180   2037  2215  26307  27446  3      4   969  bus        dynamic_debug  key-users    meminfo        scsi          timer_list
13    1679  1724  1809  2042  2216  26313  27485  30     45  970  cgroups    execdomains    keys         misc           self          tty
131   1682  1733  181   2051  2218  26622  27497  31     5   971  cmdline    filesystems    kmsg         modules        slabinfo      uptime
133   169   1741  1812  2052  2223  27     27498  32     6   972  consoles   fs             kpagecgroup  mounts         softirqs      version
135   1709  1753  1828  2053  2225  27211  27641  33     76  973  cpuinfo    interrupts     kpagecount   mtrr           stat          vmallocinfo
14    1712  1764  1831  2069  2232  27228  27664  34     77  974  crypto     iomem          kpageflags   net            swaps         vmstat
[root@ip-172-31-7-96 proc]#
```

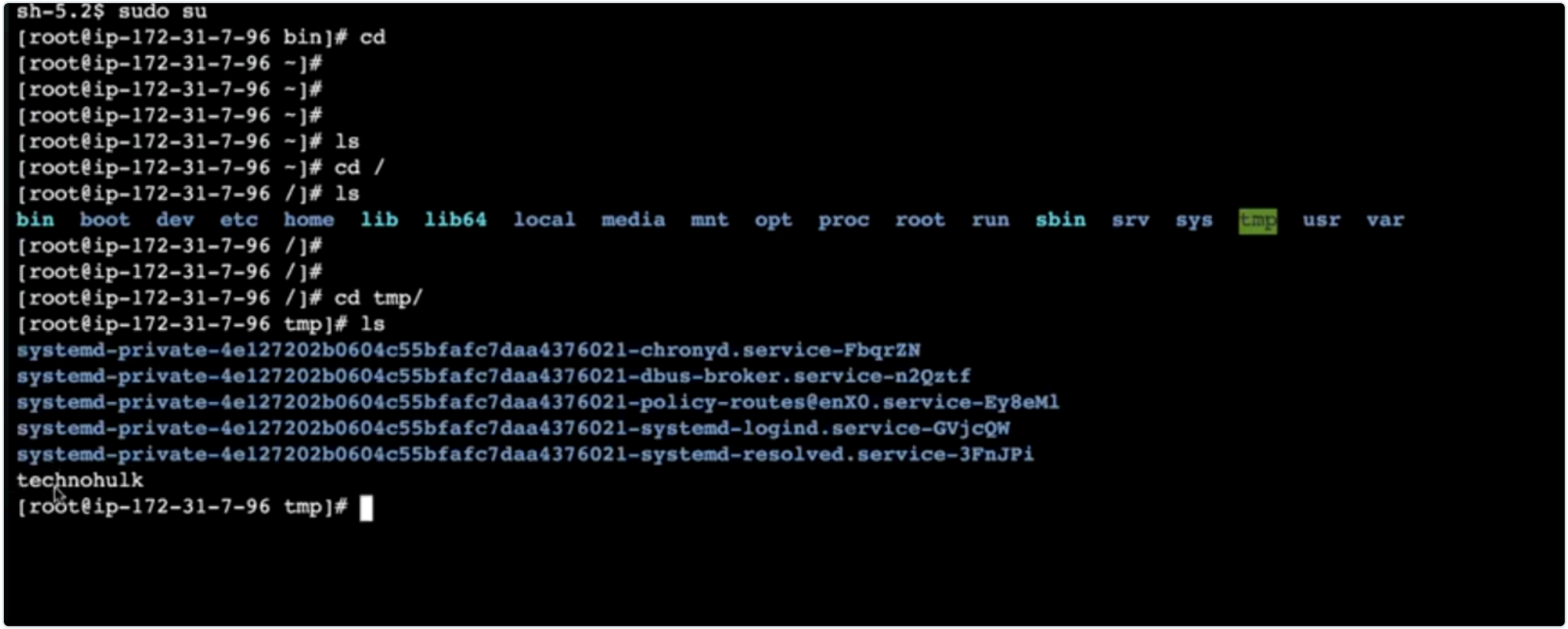
```
[root@ip-172-31-7-96 proc]# cd 27641
[root@ip-172-31-7-96 27641]# ls
arch_status  clear_refs       cwd      gid_map   map_files  mountstats  oom_score      projid_map  setgroups     statm           timers
attr         cmdline          environ  io        maps       net         oom_score_adj  root        smaps         status          timerslack_ns
autogroup    comm             exe      latency   mem        ns          pagemap        sched       smaps_rollup  syscall         uid_map
auxv         coredump_filter  fd       limits    mountinfo  numa_maps   patch_state    schedstat   stack         task            wchan
cgroup       cpuset           fdinfo   loginuid  mounts     oom_adj     personality    sessionid   stat          timens_offsets
[root@ip-172-31-7-96 27641]#
```

`cd /proc/27641/root` — a full `/` (the shim is a host process, so this is the host root)


```
[root@ip-172-31-7-96 proc]# ls
1     15    1713  1765  1837  2078  2233  27329  27714  35  79  975       devices       ioports      latency_stats  pagetypeinfo  sys           xen
10    157   1714  1767  1839  21    23    27388  27718  36  8   976       diskstats     irq          loadavg        partitions    sysrq-trigger zoneinfo
1040  158   1718  1790  2     2193  24    27444  28     37  93  acpi      dma           kallsyms     locks          pressure      sysvipc
11    16    1719  18    20    22    26298 27445  29     38  94  buddyinfo driver        kcore        mdstat         schedstat     thread-self
12    1666  1720  180   2037  2215  26307 27446  3      4   969 bus       dynamic_debug key-users    meminfo        scsi          timer_list
13    1679  1724  1809  2042  2216  26313 27485  30     45  970 cgroups   execdomains   keys         misc           self          tty
131   1682  1733  181   2051  2218  26622 27497  31     5   971 cmdline   filesystems   kmsg         modules        slabinfo      uptime
133   169   1741  1812  2052  2223  27    27498  32     6   972 consoles  fs            kpagecgroup  mounts         softirqs      version
135   1709  1753  1828  2053  2225  27211 27641  33     76  973 cpuinfo   interrupts    kpagecount   mtrr           stat          vmallocinfo
14    1712  1764  1831  2069  2232  27228 27664  34     77  974 crypto    iomem         kpageflags   net            swaps         vmstat
[root@ip-172-31-7-96 proc]# cd 27641
[root@ip-172-31-7-96 27641]# ls
arch_status  clear_refs       cwd      gid_map   map_files  mountstats  oom_score      projid_map  setgroups     statm          timers
attr         cmdline          environ  io        maps       net         oom_score_adj  root        smaps         status         timerslack_ns
autogroup    comm             exe      latency   mem        ns          pagemap        sched       smaps_rollup  syscall        uid_map
auxv         coredump_filter  fd       limits    mountinfo  numa_maps   patch_state    schedstat   stack         task           wchan
cgroup       cpuset           fdinfo   loginuid  mounts     oom_adj     personality    sessionid   stat          timens_offsets
[root@ip-172-31-7-96 27641]# cd root
[root@ip-172-31-7-96 root]# ls
bin  boot  dev  etc  home  lib  lib64  local  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@ip-172-31-7-96 root]#
```

```
[root@ip-172-31-7-96 27641]# cd root
[root@ip-172-31-7-96 root]# ls
bin  boot  dev  etc  home  lib  lib64  local  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@ip-172-31-7-96 root]#
[root@ip-172-31-7-96 root]#
[root@ip-172-31-7-96 root]#
[root@ip-172-31-7-96 root]#
[root@ip-172-31-7-96 root]# cd tmp/
[root@ip-172-31-7-96 tmp]#
[root@ip-172-31-7-96 tmp]#
[root@ip-172-31-7-96 tmp]# ls
systemd-private-4e127202b0604c55bfafc7daa4376021-chronyd.service-FbqrZN
systemd-private-4e127202b0604c55bfafc7daa4376021-dbus-broker.service-n2Qztf
systemd-private-4e127202b0604c55bfafc7daa4376021-policy-routes@enX0.service-Ey8eMl
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-logind.service-GVjcQW
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-resolved.service-3FnJPi
[root@ip-172-31-7-96 tmp]# touch technohulk
[root@ip-172-31-7-96 tmp]# pwd
/proc/27641/root/tmp
[root@ip-172-31-7-96 tmp]#
```


/proc/27641/root/tmp
[root@ip-172-31-7-96 tmp]# ls
systemd-private-4e127202b0604c55bfafc7daa4376021-chronyd.service-FbqrZN
systemd-private-4e127202b0604c55bfafc7daa4376021-dbus-broker.service-n2Qztf
systemd-private-4e127202b0604c55bfafc7daa4376021-policy-routes@enX0.service-Ey8eMl
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-logind.service-GVjcQW
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-resolved.service-3FnJPi
technohulk
[root@ip-17-31-7-96 tmp]#


sh-5.2$ sudo su
[root@ip-172-31-7-96 bin]# cd
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]#
[root@ip-172-31-7-96 ~]# ls
[root@ip-172-31-7-96 ~]# cd /
[root@ip-172-31-7-96 /]# ls
bin boot dev etc home lib lib64 local media mnt opt proc root run sbin srv sys tmp usr var
[root@ip-172-31-7-96 /]#
[root@ip-172-31-7-96 /]#
[root@ip-172-31-7-96 /]# cd tmp/
[root@ip-172-31-7-96 tmp]# ls
systemd-private-4e127202b0604c55bfafc7daa4376021-chronyd.service-FbqrZN
systemd-private-4e127202b0604c55bfafc7daa4376021-dbus-broker.service-n2Qztf
systemd-private-4e127202b0604c55bfafc7daa4376021-policy-routes@enX0.service-Ey8eMl
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-logind.service-GVjcQW
systemd-private-4e127202b0604c55bfafc7daa4376021-systemd-resolved.service-3FnJPi
technohulk
[root@ip-172-31-7-96 tmp]#